

CE 310 — Week 7 Lab: Correlation — Covariance, Pearson r, and Scatter Plots

Dataset: EnergyPlus DOE Reference Medium Office — Tucson TMY3, 2023 — Week5_DOE_MedOffice_ZoneTemp_2023.csv (8,760 rows) Topic: CORREL · COVARIANCE.S · r^2 · conditional correlation · scatter plots

Before You Begin

- Download Week5_DOE_MedOffice_ZoneTemp_2023.csv from D2L before lab (8,760 rows, one per hour).
- Download the Week7_Answer_Sheet.xlsx from D2L before lab. All answers go there.
- Column reference: G = Zone_Temp_F, H = Outdoor_Temp_F, I = Cooling_kWh.
- Verify your row count: =COUNTA(A2:A8761) should return 8760.

Answer Sheet column map:

Column	What to enter
Col A	(pre-filled question # — do not edit)
Col B	(pre-filled question text — do not edit)
Col C	Your Excel formula or method description
Col D	★ Intermediate value (★ questions only — leave blank otherwise)
Col E	Your numerical answer
Col F	Unit (pre-filled — do not edit)

Your answer cell matches the question label — A1 in row A1, B3 in row B3.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Excel Quick Reference

Task	Formula
Pearson r (all rows)	=CORREL(H2:H8761, I2:I8761)

Task	Formula
Sample covariance	=COVARIANCE.S(H2:H8761, I2:I8761)
Conditional r — FILTER	=CORREL(FILTER(H:H, E:E="Summer"), FILTER(I:I, E:E="Summer"))
Conditional r — Array (Ctrl+Shift+Enter)	{=CORREL(IF(E2:E8761="Summer", H2:H8761), IF(E2:E8761="Summer", I2:I8761))}
Count subgroup size	=COUNTIF(E2:E8761,"Summer") or =COUNTIF(F2:F8761,"Yes")
r^2 (coefficient of determination)	= r_cell ^ 2
Standard deviation (sample)	=STDEV(I2:I8761)
Verify: $r = \text{cov} / (\sigma_1 \times \sigma_2)$	= COVARIANCE.S(H2:H8761,I2:I8761) / (STDEV(H2:H8761) * STDEV(I2:I8761)) should equal CORREL

Background

Today you'll quantify how strongly each variable moves with cooling energy.

Setup

1. Open Week5_DOE_MedOffice_ZoneTemp_2023.csv in Excel.
2. Verify 8,760 data rows: =COUNTA(A2:A8761) → should return 8760.
3. Confirm column letters: A = Date, B = Month, C = Hour, D = DayType, E = Season, F = BusinessHours, G = Zone_Temp_F, H = Outdoor_Temp_F, I = Cooling_kWh.
4. Open Week7_Answer_Sheet.xlsx. Enter your name and NetID.

Section A — Introduction to CORREL (8 questions × 2 pts = 16 pts)

Each question uses CORREL or COVARIANCE.S on the full 8,760-row dataset, or simple arithmetic on those results. Enter answers rounded to 4 decimal places.

Formula syntax: - =CORREL(array1, array2) — returns Pearson r between two arrays. - =COVARIANCE.S(array1, array2) — returns sample covariance (ddof = 1).

A1 (2 pts): Compute $r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for all 8,760 hours.

Interpretation note: This correlation is moderate-to-strong. But notice the key insight hiding behind it: the 5.8× compression ratio in this dataset ($\sigma_{\text{outdoor}} / \sigma_{\text{zone}} \approx 16.92 /$

2.92) shows how tightly the HVAC controls zone temperature relative to outdoor swings. The moderate-to-strong correlation tells you there is a meaningful relationship — when outdoor temperature rises, zone temperature tends to rise too — but the HVAC system strongly limits how much it rises. Correlation describes the direction and linearity; the compression ratio tells you the magnitude of control.

A2 (2 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for all 8,760 hours.

This is the main pair for this lab. A strong positive correlation — hotter outdoor hours tend to have higher cooling loads.

A3 (2 pts): Compute $r(\text{Zone_Temp_F}, \text{Cooling_kWh})$ for all 8,760 hours.

This is the weak correlation. Zone temperature is tightly controlled at the thermostat set-point — it doesn't vary much ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$), so it carries little information about how hard the cooling system is working. Outdoor temperature ($\sigma_{\text{outdoor}} \approx 16.9^\circ\text{F}$) varies much more and is the dominant driver of cooling load.

A4 (2 pts — multiple choice): Which correlation from A1–A3 has the largest absolute value?

(A) $r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F}) = 0.7529$ (B) $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh}) = 0.7752$ (C) $r(\text{Zone_Temp_F}, \text{Cooling_kWh}) = 0.2764$ (D) All three are equal — CORREL always returns the same value

A5 (2 pts): Compute $r^2(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — the coefficient of determination.

r^2 gives the percentage of variance in Cooling_kWh linearly explained by Outdoor_Temp_F across all 8,760 hours. The remaining unexplained variance is driven by other factors — occupancy, internal equipment loads, solar gain, humidity, and HVAC scheduling.

A6 (2 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for business hours only (BusinessHours = "Yes").

Note: During occupied hours, the outdoor-cooling relationship strengthens substantially compared to the all-hours result.

A7 (2 pts — multiple choice): $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh}) \approx 0.78$ for all hours. Which interpretation is correct?

(A) Outdoor temperature causes cooling load. (B) There is a strong positive linear relationship — hotter outdoor hours tend to have higher cooling loads. (C) 78% of cooling hours occurred above the outdoor mean temperature. (D) The standard deviation of Cooling_kWh equals 0.78 times the standard deviation of Outdoor_Temp_F.

Causation taxonomy (ungraded): The lecture identified three reasons correlation does not equal causation: confounders (a third variable drives both), reverse causation (Y causes X rather than X causing Y), and coincidence. For this pair, outdoor temperature is a plausible physical driver of cooling load — but Season is also a confounder that independently

drives both high outdoor temperatures and high cooling demand. Whenever you report a correlation, state which of the three pitfalls you have ruled out and why.

A8 (2 pts — multiple choice): $r(\text{Zone_Temp_F}, \text{Cooling_kWh}) \approx 0.28$ from A3 — much lower than $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh}) \approx 0.78$ from A2. The most likely engineering explanation is:

(A) Outdoor temperature has no effect on zone temperature. (B) Cooling energy is measured in kWh, which is incompatible with °F in CORREL. (C) Zone temperature is tightly controlled near the setpoint ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$), so it contains little information about the cooling load — all the variation in cooling demand shows up in outdoor temperature, not in the nearly-constant zone temperature. (D) Zone temperature and cooling energy are negatively correlated.

Section B Core — Seasonal and Conditional Correlations (B1–B6)

All r values → 4 decimal places.

B1 ★ (5 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for Summer hours only.

Col D: number of Summer hours.

B2 (4 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for Winter hours only.

Note: Winter still shows a moderate positive correlation. Even in winter, on warmer winter days ($50\text{--}65^\circ\text{F}$), some cooling is needed — primarily during business hours when internal gains from equipment and people build up.

B3 (4 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for Spring hours only.

Spring has the strongest outdoor-cooling correlation of the four seasons.

B4 (4 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for Fall hours only.

B5 (4 pts): Compute the range of seasonal correlations: $\text{MAX}(B1, B2, B3, B4) - \text{MIN}(B1, B2, B3, B4)$.

This small range shows that the outdoor-cooling linear relationship is remarkably consistent across all four seasons. The relationship is strong year-round — not just in summer.

B6 ★ (5 pts): Compute $\text{COVARIANCE.S}(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for all 8,760 hours.

Col D: $\sigma(\text{Cooling_kWh})$ using STDEV.

Covariance has units ($^\circ\text{F} \times \text{kWh}$); correlation is dimensionless. Verify: $\text{cov} / (\sigma_{\text{outdoor}} \times \sigma_{\text{cooling}})$ should equal your r from A2 (this verification is stated for understanding; it is not a separate graded answer).

Engineering Context: ASHRAE 90.1-2022 requires cooling equipment sized for peak load. Correlation analysis identifies which variables are dominant drivers of that peak — informing whether a simple or complex sizing model is appropriate.

Section B Extension — Business Hours, r^2 , and Model Diagnostics (B7–B14)

B7 ★ (5 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for business hours (`BusinessHours = "Yes"`).

Col D: number of business hours.

Note: This computes the same quantity as A6. During business hours, outdoor temperature is an almost perfect linear predictor of cooling load.

B8 (4 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for non-business hours (`BusinessHours = "No"`).

During non-business hours, cooling is reduced (setback mode or off entirely) so the outdoor-cooling relationship weakens.

B9 (4 pts): Compute r^2 for business hours: square your B7 answer.

During occupied hours, outdoor temperature explains a very high proportion of the variance in cooling energy. The remaining unexplained variance represents other factors: solar gain variability, differences in internal loads from day to day, and HVAC control dynamics.

B10 (4 pts): Compute $r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for business hours only.

Compare to A1 (all-hours result). During business hours, zone temperature actually tracks outdoor temperature MORE closely — perhaps surprising at first. The reason: during business hours, the HVAC responds dynamically to outdoor conditions, creating a structured relationship. The zone temperature rises and falls with outdoor temperature within the narrow band the thermostat allows. All-hours r is lower because unoccupied setback periods add noise.

B11 (4 pts): Compute $r(\text{Month}, \text{Outdoor_Temp_F})$ for all 8,760 hours.

Month is numeric 1–12. The low correlation occurs because outdoor temperature in Tucson follows a non-linear seasonal curve — temperatures peak in July (Month 7), then fall. Month is a poor linear predictor of outdoor temperature because the relationship is parabolic (rises then falls), not monotone. A linear correlation cannot capture this curved pattern.

B12 (4 pts): Compute $r(\text{Month}, \text{Cooling_kWh})$ for all 8,760 hours.

Target shape for Chart 1

Figure 1: Target shape for Chart 1

Even lower than B11. Month is a nearly useless linear predictor of cooling energy — the cooling-vs-month relationship is also parabolic. Compare to $r(\text{Outdoor}, \text{Cooling})$: Month is a calendar label, while outdoor temperature is the physical mechanism. Use the mechanism, not the calendar, in your regression model.

B13 ★ (5 pts): Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for non-zero cooling hours only ($\text{Cooling_kWh} > 0$).

Col D: hours where $\text{Cooling_kWh} > 0$ (COUNTIF).

Note: This is slightly higher than the all-hours r . The zero-cooling hours are mostly nights and winter hours at low outdoor temperatures — they anchor both variables at their low ends and modestly inflate the all-hours correlation. Removing them gives a cleaner picture of the outdoor-cooling relationship during actual cooling operation.

B14 (4 pts): Compute $\text{COVARIANCE.S}(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for all 8,760 hours.

Compare to B6. The zone-outdoor covariance is much smaller — not because the relationship is weaker (r values are similar for both pairs), but because Zone_Temp_F has a much smaller standard deviation ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$) than Cooling_kWh ($\sigma_{\text{cooling}} \approx 25.3$ kWh). Covariance depends on the scale of both variables. Correlation normalizes this: $r = \text{cov} / (\sigma_1 \times \sigma_2)$.

Required Charts

Insert both charts in your working Excel file.

Chart 1 (7 pts) — Scatter: Outdoor Temperature vs. Cooling Energy

Plot Outdoor_Temp_F (x) vs. Cooling_kWh (y) for all 8,760 rows as dots only (Insert → Scatter → dots only). Add a linear trendline with “Display R-squared value on chart” checked.

- Title: Outdoor Temperature vs. Cooling Energy — Tucson TMY3 (8,760 hours)
- Axes: Outdoor Temperature ($^\circ\text{F}$) on x, Cooling Energy (kWh) on y
- The R^2 on the chart should match your A5 answer (± 0.01)

Chart 2 (8 pts) — Seasonal Scatter: Outdoor Temperature vs. Cooling Energy

Plot Outdoor_Temp_F vs. Cooling_kWh as four separate series (one per season) in different colors. Label each series in the legend with the season name and its r value from B1–B4.

Target shape for Chart 2

Figure 2: Target shape for Chart 2

- Title: Seasonal Correlation: Outdoor Temperature vs. Cooling Energy
 - Axes: Outdoor Temperature (°F) on x, Cooling Energy (kWh) on y
-

Memo Deliverable (9 pts)

Type your memo directly in the Written sheet of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed. Address the memo to: “Building Operations Team, DOE Reference Medium Office Building, Tucson AZ.”

Paragraph 1 (3 pts): Report your computed values for A1, A2, and A3. Which pair has the strongest correlation? Explain why $r(\text{Zone_Temp_F}, \text{Cooling_kWh})$ is weak, referencing the $5.8\times$ compression ratio in this dataset ($\sigma_{\text{outdoor}} \approx 16.92^\circ\text{F}$ versus $\sigma_{\text{zone}} \approx 2.92^\circ\text{F}$): zone temperature is so tightly controlled by the HVAC ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$) that it carries little signal about how hard the cooling system is working.

Paragraph 2 (3 pts): Report the four seasonal r values from B1–B4. Which season has the strongest outdoor-cooling correlation and why? Reference the ASHRAE 90.1-2022 peak-load sizing requirement from the Engineering Context note above: the outdoor-cooling relationship is strong in all seasons. Discuss what this consistency means for equipment sizing.

Paragraph 3 (3 pts) — ArcE focus: Report your computed values for B7, B8, and B9. How does the outdoor-cooling correlation differ between business hours and non-business hours? What does your $r^2(\text{business hours})$ result tell an architect about occupied-mode cooling energy? What does the remaining $(1 - r^2)$ represent, and what design factors might explain it?

Submission: Save as CE310_W7_<NetID>.xlsx and upload to the Week 7 D2L dropbox. A 5-point deduction applies if your file cannot be matched to your student record.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
- ☐ File saved as .xlsx and named CE310_W7_<NetID>.xlsx
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit

- ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in Col D
 - ☐ Uploaded to D2L before the deadline
-

Grading Summary

Component	Points
Section A (A1–A8)	16
Section B (B1–B14)	60
Chart 1 (scatter + trendline)	7
Chart 2 (seasonal scatter)	8
Memo (3 paragraphs)	9
Total	100

Point structure: Section A = 8×2.0 pts (0.5 approach + 1.5 answer each). Section B = 10 standard $\times 4.0$ pts + 4 intermediate ★ $\times 5.0$ pts = 40 + 20 = 60 pts. ★ questions: B1, B6, B7, B13.